CLIENT

AXIATA (BANGLADESH) LTD.

BORELOG FOR

THE PROPOSED TELECOMMUNICATION TOWER AT SODURKHIL, MANIKCHARI, KHAGRACHARI.

(AUGUST, 2010)

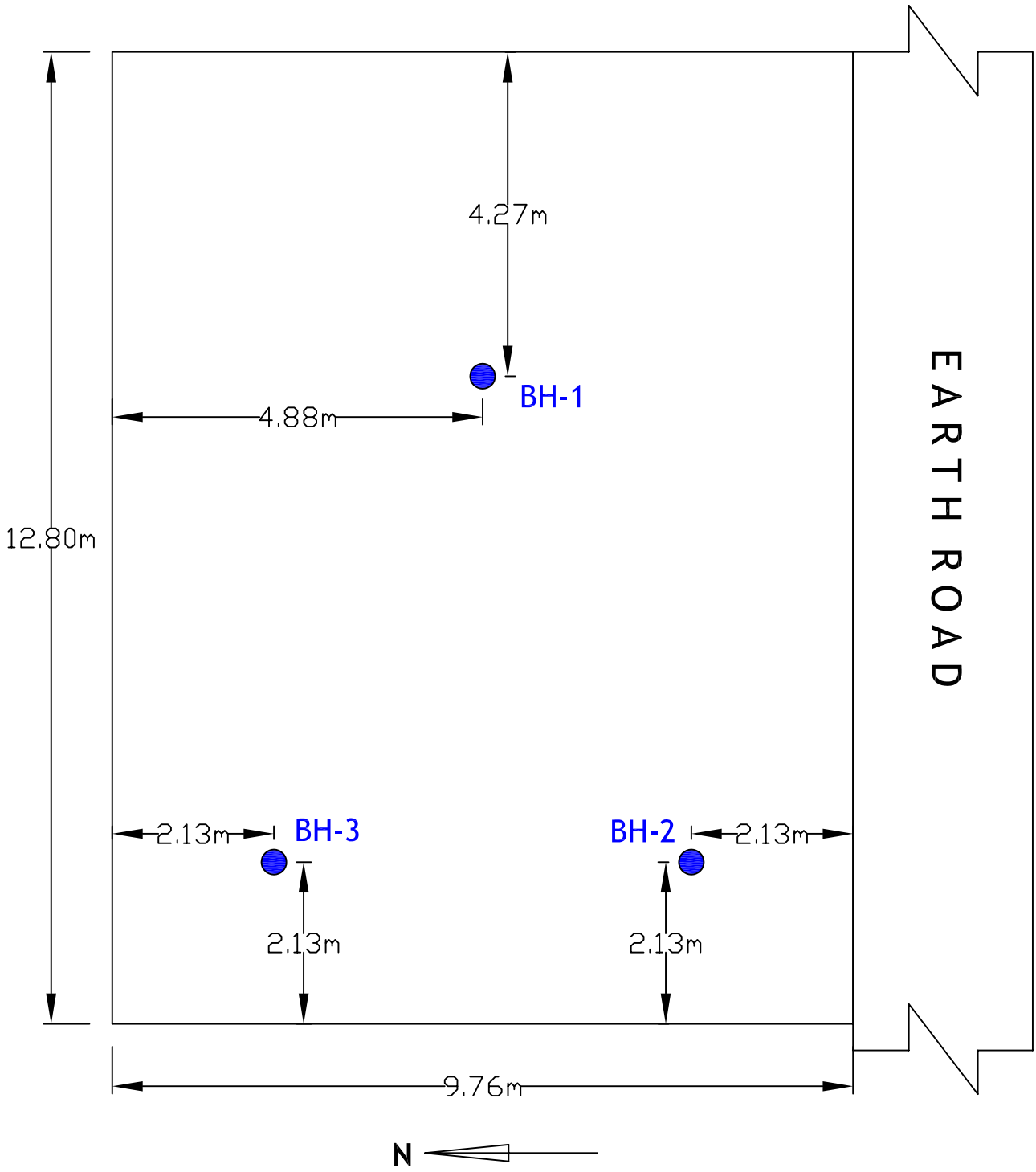
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A1-LOCATION MAP

CLIENT: AXIATA (BANGLADESH) LTD.

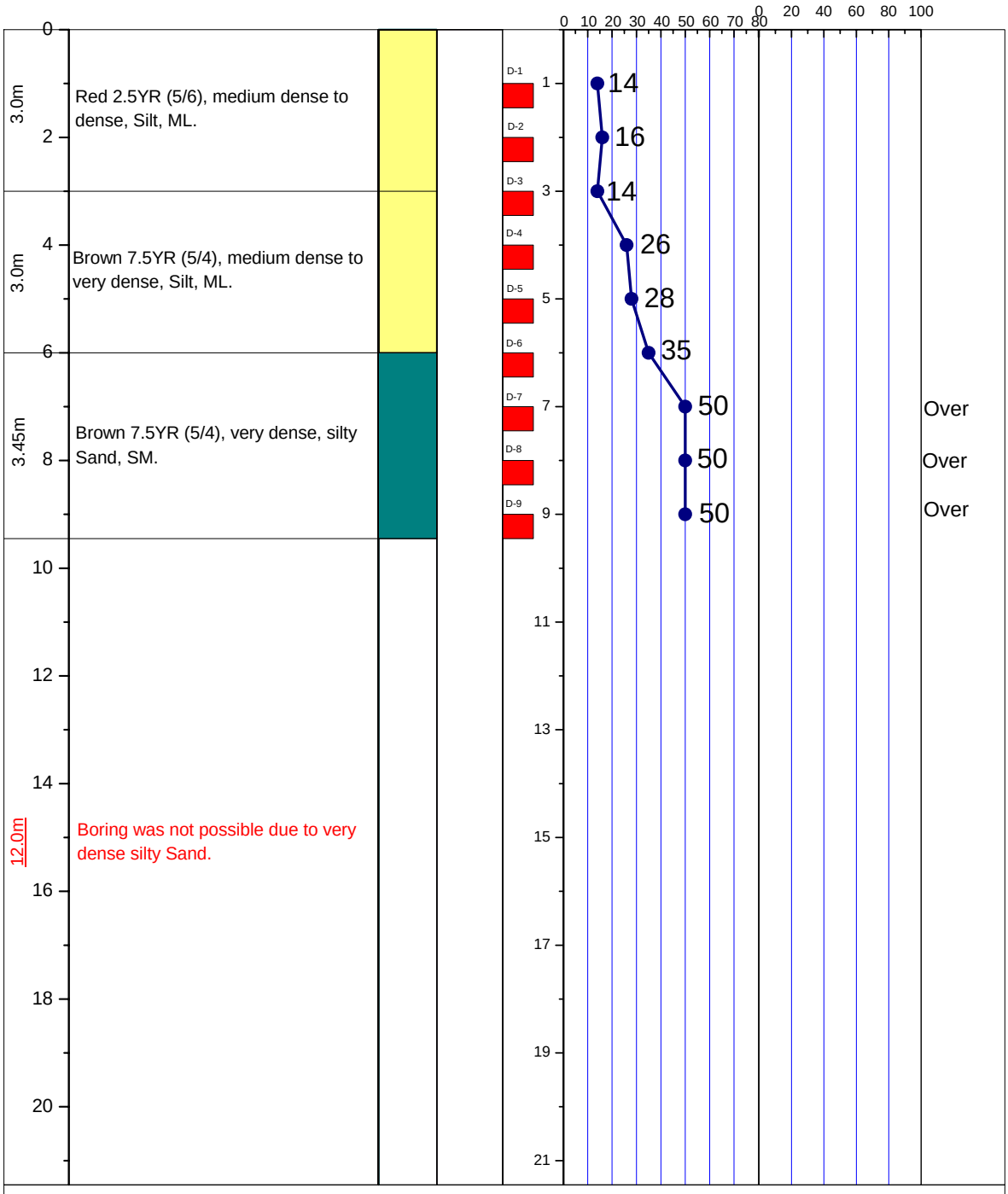
SHOWING BORING LOCATION

LOCATION: SODURKHIL, MANIKCHARI, KHAGRACHARI.



PROSOIL FOUNDATION CONSULTANT

A2	PROSOIL	BOREHOLE NO: BH-1	
		LOCATION: Sodurkhil, Manikchari, khagrachari.	
PROJECT : Telecommunication Tower.		DEPTH OF BORING: 9.0m	R. L.: 0.46m above from Road Level.
CLIENT : AXIATA (Bangladesh) Ltd.		DATE OF BORING: 02-08-2010.	G. W. T : 3.66m below from E. G. L.



LEGEND FOR BORELOG AND GRAIN SIZE DISTRIBUTION (BASED ON 'ASTM CLASSIFICATION D-2487')

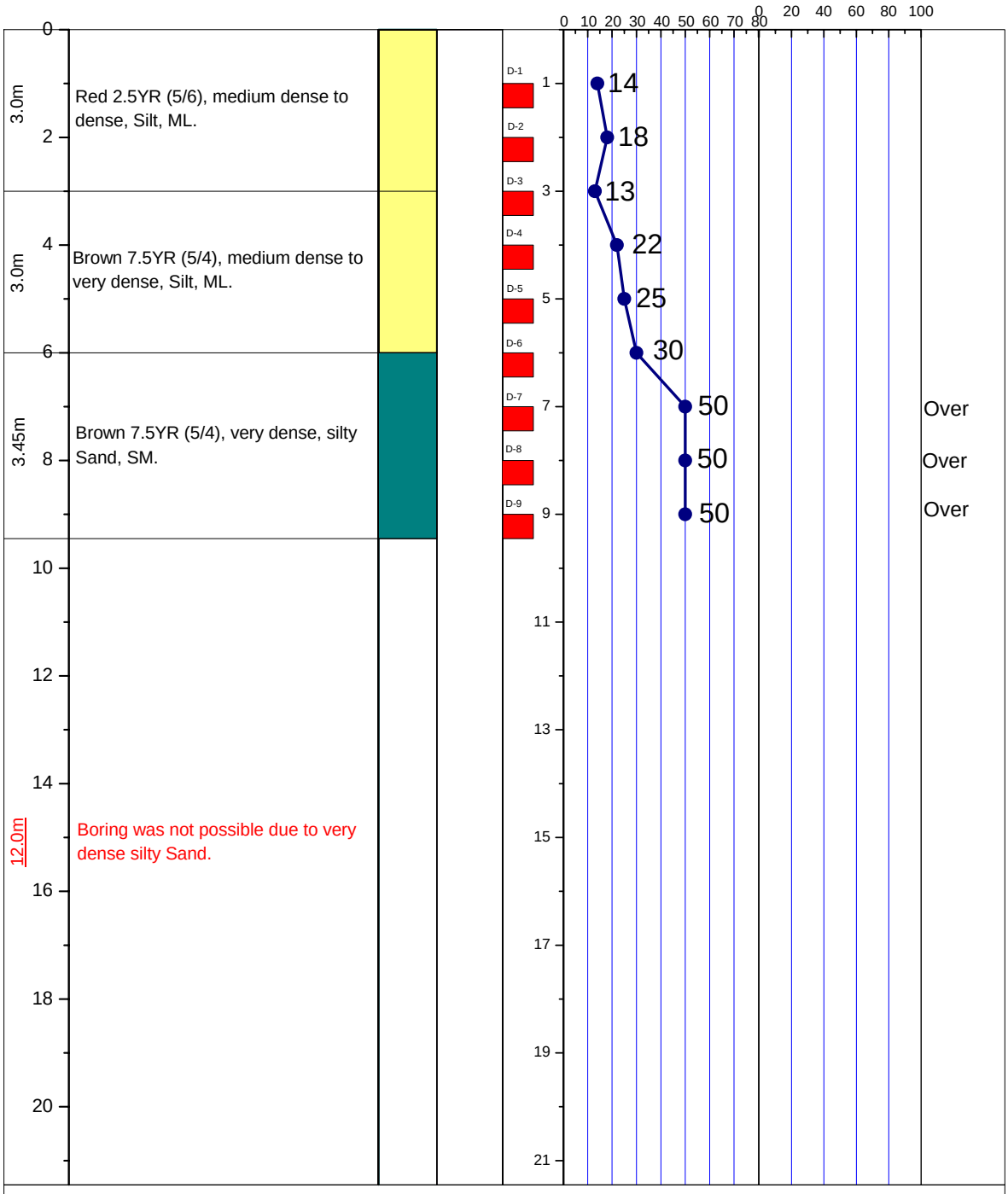
PLASTIC CLAY	SAND	SILT	SHALE
PEAT/HIGHLY ORGANIC SOIL	MICACEOUS FINE SAND/SILT	UNDISTURBED SAMPLE	DISTURBED SAMPLE

N- BLOWS DELIVERED PER FOOT BY A 140 LB HAMMER FALLING 30 INCHES

MAX^{m} PENETRATION= 50 BLOWS

AFTER 50 BLOWS PENETRATION DEPTH WERE MEASURED

A2	PROSOIL	BOREHOLE NO: BH-2	
		LOCATION: Sodurkhil, Manikchari, khagrachari.	
PROJECT : Telecommunication Tower.		DEPTH OF BORING: 9.0m	R. L.: 0.46m above from Road Level.
CLIENT : AXIATA (Bangladesh) Ltd.		DATE OF BORING: 02-08-2010.	G. W. T : 3.66m below from E. G. L.



LEGEND FOR BORELOG AND GRAIN SIZE DISTRIBUTION (BASED ON 'ASTM CLASSIFICATION D-2487')

PLASTIC CLAY	SAND	SILT	SHALE
PEAT/HIGHLY ORGANIC SOIL	MICACEOUS FINE SAND/SILT	UNDISTURBED SAMPLE	DISTURBED SAMPLE

N- BLOWS DELIVERED PER FOOT BY A 140 LB HAMMER FALLING 30 INCHES

MAX^{m} PENETRATION= 50 BLOWS

AFTER 50 BLOWS PENETRATION DEPTH WERE MEASURED

A2	PROSOIL	BOREHOLE NO: BH-3	
		LOCATION: Sodurkhil, Manikchari, khagrachari.	
PROJECT : Telecommunication Tower.		DEPTH OF BORING: 9.0m	R. L.: 0.46m above from Road Level.
CLIENT : AXIATA (Bangladesh) Ltd.		DATE OF BORING: 03-08-2010.	G. W. T : 3.66m below from E. G. L.

THICKNESS	SOIL STRATA	BORELOG	UNDISTURBED SAMPLE	DISTURBED SAMPLE	DEPTH (m) SPT VALUES (BLOWS / 300mm)	Grain Size Distribution, % Clay %, Silt %, Sand %	Max^{m} Penetration
3.0m	Red 2.5YR (5/6), dense to medium dense, Silt, ML.			D-1 D-2 D-3	15 15 13		
3.0m	Brown 7.5YR (5/4), medium dense to very dense, Silt, ML.			D-4 D-5	25 31		
3.45m	Brown 7.5YR (5/4), very dense, silty Sand, SM.			D-6 D-7 D-8 D-9	37 50 50 50		Over Over Over
12.0m	Boring was not possible due to very dense silty Sand.						

LEGEND FOR BORELOG AND GRAIN SIZE DISTRIBUTION (BASED ON 'ASTM CLASSIFICATION D-2487')

PLASTIC CLAY	SAND	SILT	SHALE
PEAT/HIGHLY ORGANIC SOIL	MICACEOUS FINE SAND/SILT	UNDISTURBED SAMPLE	DISTURBED SAMPLE

N- BLOWS DELIVERED PER FOOT BY A 140 LB HAMMER FALLING 30 INCHES

MAX^{m} PENETRATION= 50 BLOWS

AFTER 50 BLOWS PENETRATION DEPTH WERE MEASURED

A5: (TABLE - 2) SKIN FRICTION & END-BEARING

PILE SIZE 1.667 FT F.S. 3 upto 7.0m 3 below 7.0m
CROSS SECTIONAL AREA 2.18 SFT
BH NO 1

CUM. DEPTH IN METER	DEFTH IN FT.	CUM. DEPTH IN FT	S.P.T. VALUE	CORRECTED S.P.T. VALUE N-(N-15)/2	AVERAGE SPT VALUE	SURFACE AREA IN PILE (SFT)	TOTAL SURFACE AREA IN PILE (SFT)	SKIN FRICTION (TONS) ALLOWABLE Qa=.02N/F.S. TSF	TOTAL SKIN FRICTION (TONS) ALLOWABLE Qa=.02N/F.S. TSF	END BEARING CAPACITY(TONS) ALLOWABLE Qa=4N/F.S. TSF	WORKING LOAD OF PILE (TONS)	SKIN FRICTION (TSF) ALLOWABLE Qa=.02N/F.S. TSF	END BEARING CAPACITY(TSF) Qa=4N/F.S. TSF
1	3.281	3.28	14	14	7	17.186	17.18619	0.802022	0.802022	40.74	41.54	0.046667	18.66667
2	3.281	6.56	16	15.5	14.75	17.186	34.37238	1.689975	2.491998	45.11	47.60	0.098333	20.66667
3	3.281	9.84	14	14	14.75	17.186	51.55857	1.689975	4.181973	40.74	44.92	0.098333	18.66667
4	3.281	13.1	26	20.5	17.25	17.186	68.74476	1.976412	6.158385	59.66	65.81	0.115	27.33333
5	3.281	16.4	28	21.5	21	17.186	85.93095	2.406067	8.564452	62.57	71.13	0.14	28.66667
6	3.281	19.7	35	25	23.25	17.186	103.1171	2.66386	11.22831	72.75	83.98	0.155	33.33333
7	3.281	23	50	32.5	28.75	17.186	120.3033	3.29402	14.52233	94.58	109.10	0.191667	43.33333
8	3.281	26.2	50	32.5	32.5	17.186	137.4895	3.723675	18.24601	94.58	112.82	0.216667	43.33333
9	3.281	29.5	50	32.5	32.5	17.186	154.6757	3.723675	21.96968	94.58	116.55	0.216667	43.33333

A5: (TABLE - 2) SKIN FRICTION & END-BEARING

PILE SIZE 1.667 FT F.S. 3 upto 7.0m 3 below 7.0m
CROSS SECTIONAL AREA 2.18 SFT
BH NO 2

CUM. DEPTH IN METER	DEFTH IN FT.	CUM. DEPTH IN FT	S.P.T. VALUE	CORRECTED S.P.T. VALUE N-(N-15)/2	AVERAGE SPT VALUE	SURFACE AREA IN PILE (SFT)	TOTAL SURFACE AREA IN PILE (SFT)	SKIN FRICTION (TONS) ALLOWABLE Qa=.02N/F.S. TSF	TOTAL SKIN FRICTION (TONS) ALLOWABLE Qa=.02N/F.S. TSF	END BEARING CAPACITY(TONS) ALLOWABLE Qa=4N/F.S. TSF	WORKING LOAD OF PILE (TONS)	SKIN FRICTION (TSF) ALLOWABLE Qa=.02N/F.S. TSF	END BEARING CAPACITY(TSF) Qa=4N/F.S. TSF
1	3.281	3.28	14	14	7	17.186	17.18619	0.802022	0.802022	40.74	41.54	0.046667	18.66667
2	3.281	6.56	18	16.5	15.25	17.186	34.37238	1.747263	2.549285	48.02	50.57	0.101667	22
3	3.281	9.84	13	13	14.75	17.186	51.55857	1.689975	4.23926	37.83	42.07	0.098333	17.33333
4	3.281	13.1	22	18.5	15.75	17.186	68.74476	1.80455	6.04381	53.84	59.88	0.105	24.66667
5	3.281	16.4	25	20	19.25	17.186	85.93095	2.205561	8.249371	58.20	66.45	0.128333	26.66667
6	3.281	19.7	30	23	21.25	17.186	103.1171	2.43471	10.68408	65.48	76.16	0.141667	30
7	3.281	23	50	32.5	27.5	17.186	120.3033	3.150802	13.83488	94.58	108.41	0.183333	43.33333
8	3.281	26.2	50	32.5	32.5	17.186	137.4895	3.723675	17.55856	94.58	112.14	0.216667	43.33333
9	3.281	29.5	50	32.5	32.5	17.186	154.6757	3.723675	21.28223	94.58	115.86	0.216667	43.33333

NOTE:

a) fsa = Allowable value of skin friction (F.S. =3.0)
b) qpa = Allowable value of point bearing capacity (F. S. = 3.0)
c) The above value of fsa and qpa are for driven piles. For bored pile values may be taken as 3/4th in making preliminary estimate.
d) Final estimates of pile load carrying capacity should be confirmed by load test

A5: (TABLE - 2) SKIN FRICTION & END-BEARING

PILE SIZE 1.667 FT F.S. 3 upto 7.0m 3 below 7.0m

CROSS SECTIONAL AREA 2.18 SFT

BH NO 3

CUM. DEPTH IN METER	DEFTH IN FT.	CUM. DEPTH IN FT	S.P.T. VALUE	CORRECTED S.P.T. VALUE N-(N-15)/2	AVERAGE SPT VALUE	SURFACE AREA IN PILE (SFT)	TOTAL SURFACE AREA IN PILE (SFT)	SKIN FRICTION (TONS) ALLOWABLE Qa=.02N/F.S. TSF	TOTAL SKIN FRICTION (TONS) ALLOWABLE Qa=.02N/F.S. TSF	ALLOWABLE END BEARING CAPACITY(TONS) Qa=4N/F.S. TSF	WORKING LOAD OF PILE (TONS) ALLOWABLE	SKIN FRICTION (TSF) ALLOWABLE Qa=.02N/F.S. TSF	END BEARING CAPACITY(TSF) Qa=4N/F.S. TSF ALLOWABLE
1	3.281	3.28	15	15	7.5	17.186	17.18619	0.85931	0.85931	43.65	44.51	0.05	20
2	3.281	6.56	15	15	15	17.186	34.37238	1.718619	2.577929	43.65	46.23	0.1	20
3	3.281	9.84	13	13	14	17.186	51.55857	1.604044	4.181973	37.83	42.01	0.093333	17.33333
4	3.281	13.1	25	20	16.5	17.186	68.74476	1.890481	6.072454	58.20	64.27	0.11	26.66667
5	3.281	16.4	31	23	21.5	17.186	85.93095	2.463354	8.535808	66.93	75.47	0.143333	30.66667
6	3.281	19.7	37	26	24.5	17.186	103.1171	2.807078	11.34289	75.66	87.00	0.163333	34.66667
7	3.281	23	50	32.5	29.25	17.186	120.3033	3.351307	14.69419	94.58	109.27	0.195	43.33333
8	3.281	26.2	50	32.5	32.5	17.186	137.4895	3.723675	18.41787	94.58	112.99	0.216667	43.33333
9	3.281	29.5	50	32.5	32.5	17.186	154.6757	3.723675	22.14154	94.58	116.72	0.216667	43.33333

NOTE:

a) fsa = Allowable value of skin friction (F.S. =3.0)

b) qpa = Allowable value of point bearing capacity (F. S. = 3.0)

c) The above value of fsa and qpa are for driven piles. For bored pile values may be taken as 3/4th in making preliminary estimate.

d) Final estimates of pile load carrying capacity should be confirmed by load test